

CORNER-FREE 4-COP-WIN GRAPHS ON 20 VERTICES, AND A RETRACTION QUESTION OF TURCOTTE AND YVON

ABSTRACT. Let M_k be the least order of a connected graph with cop number exactly k . Turcotte and Yvon, who proved $M_4 = 19$, ask in the discussion of their paper whether the k -cop-win graphs on $M_k + 1$ vertices always retract onto k -cop-win graphs on M_k vertices, and single out the case $k = 4$. We answer that case negatively: two connected 4-regular graphs of girth 5 on $20 = M_4 + 1$ vertices have cop number exactly 4 and have *no corner*, hence no retract of order 19 whatsoever. They are exhibited below by their graph6 strings and their adjacency lists, and the certificate is short enough to check by hand: girth at least 5 with $\delta = 4$ gives $c \geq 4$ by a theorem of Aigner and Fromme, an explicit three-element set whose complement induces a forest gives $c \leq 4$, and girth at least 5 with $\delta \geq 2$ forbids corners. Only $k = 4$ is settled here: $k = 2$ is already false in the source, $k = 3$ is true there by its own classification, and $k \geq 5$ is untouched because M_5 is unknown.

1. THE QUESTION

We use the conventions of Turcotte and Yvon [4]. In the game of Cops and Robbers on a finite connected simple graph G , the cops first choose their vertices, then the robber chooses his; thereafter the cops move, then the robber, alternately, and on his turn a player may stay put or move to an adjacent vertex. The cops win if a cop ever occupies the robber's vertex, and $c(G)$ is the least number of cops with a winning strategy. Following [4], G is *k-cop-win* if $c(G) = k$ *exactly*, and

$$M_k = \min\{|V(G)| : G \text{ connected and } c(G) = k\}.$$

Thus $M_2 = 4$ (attained by C_4), $M_3 = 10$ (attained by the Petersen graph), and the main theorem of [4] is $M_4 = 19$, attained by the Robertson graph [3]. A graph H is a *retract* of G if H is an *induced* subgraph of G and there is a map $f: V(G) \rightarrow V(H)$ with $f|_{V(H)} = \text{id}$ such that $xy \in E(G)$ implies $f(x)f(y) \in E(H)$ or $f(x) = f(y)$.

In the discussion following their classification of the 3-cop-win graphs on 11 vertices, the authors ask the following. We quote it in full, from the arXiv e-print 2006.02998v2 of [4].

“This is an interesting phenomenon: the unique 3-cop-win graph on 10 vertices is a retract of all the 3-cop-win graphs

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on 11 vertices. This behaviour does not occur for the 2-cop-win graphs: the minimum 2-cop-win graph is the 4-cycle, on which the 5-cycle does not retract. Although we will not have any answer for this question in this article, it would be interesting to know whether in general (even for 4-cop-win graphs only), the k -cop-win graphs on $M_k + 1$ vertices can be retracted on k -cop-win graph(s) on M_k vertices.”

This is an unnumbered question in running prose, and not the numbered Conjecture of [4], which is a different statement – that no connected graph of order 19 with maximum degree 5 or 6 has cop number 4 – and is not addressed here; in particular we do not claim, and [4] does not prove, that the Robertson graph is the only 4-cop-win graph on 19 vertices. The authors state that $k = 2$ already fails, their witness being C_5 , which does not retract onto C_4 ; and $k = 3$ is true, since the classification just cited shows that all six 3-cop-win graphs on 11 vertices retract onto the Petersen graph. We settle the case the authors name, $k = 4$, in the negative. Since $M_4 = 19$, the relevant order is $M_4 + 1 = 20$.

2. THE TWO GRAPHS

Let W_1 and W_2 be the graphs on the vertex set $\{0, 1, \dots, 19\}$ given by the following graph6 strings, one 33-byte line each, $33 = 1 + \lceil \binom{20}{2}/6 \rceil = 1 + 32$:

W1 = S????B?K?W?Wig`cIP?j?QSA`GCoGBGG?
W2 = S???CA?_c?o@_Q_i0AcU@OK`@c0?M0?Bg?

Equivalently, by their adjacency lists (the label v followed by the four neighbours of v):

W1	W2
0:8,13,17,18	0:6,9,13,15
1:8,14,16,19	1:7,12,15,17
2:9,12,17,19	2:8,13,16,17
3:9,14,15,18	3:9,10,14,16
4:10,12,16,18	4:10,11,12,13
5:10,13,15,19	5:11,14,17,18
6:11,12,13,14	6:0,12,16,18
7:11,15,16,17	7:1,13,18,19
8:0,1,12,15	8:2,14,15,19
9:2,3,13,16	9:0,3,17,19
10:4,5,14,17	10:3,4,15,18
11:6,7,18,19	11:4,5,16,19
12:2,4,6,8	12:1,4,6,14
13:0,5,6,9	13:0,2,4,7
14:1,3,6,10	14:3,5,8,12
15:3,5,7,8	15:0,1,8,10
16:1,4,7,9	16:2,3,6,11
17:0,2,7,10	17:1,2,5,9
18:0,3,4,11	18:5,6,7,10
19:1,2,5,11	19:7,8,9,11

Each is connected, 4-regular on 20 vertices with 40 edges, and of girth 5; the 5-cycles 08153180 in W_1 and 06121150 in W_2 show the girth is not more than 5, and the absence of triangles together with $\max_{a \neq b} |N(a) \cap N(b)| = 1$ shows it is not less.

Neither graph is new here: they are members of Meringer’s census [2] of connected 4-regular graphs of girth at least 5, being Graph 2 and Graph 1 of the file 20_4_5.asc, and their automorphism groups have orders 96 and 20 respectively, which is how the identification is checked here.

3. THE CERTIFICATE

Write $N[S] = S \cup \bigcup_{v \in S} N(v)$. A vertex x is a *corner* (or is *dominated*) if there is $u \neq x$ with $N(x) \subseteq N[u]$.

Theorem 3.1 (Aigner and Fromme [1]). *If G has girth at least 5, then $c(G) \geq \delta(G)$.*

Lemma 3.2. *Let G be connected and suppose some $S \subseteq V(G)$ with $|S| = 3$ has $G[V(G) \setminus N[S]]$ a forest. Then $c(G) \leq 4$.*

Proof. Place three cops on S and let them never move. The robber cannot choose a vertex of $N[S]$, since a cop would step onto him at once, and for the same reason he can never enter $N[S]$ later; so he is confined for the whole game to $R = V(G) \setminus N[S]$, and in fact to the component T of $G[R]$ containing his initial vertex. As $G[R]$ is a forest, T is a tree. A fourth cop walks to T – possible since G is connected – and then plays the one-cop winning strategy on the tree T , in which the robber’s legal moves are exactly the edges of T . Hence four cops suffice. $\square$

This argument is not ours: it is the one [4] applies to the Robertson graph, and is attributed there to earlier work of Beveridge et al.

Lemma 3.3. *If G has girth at least 5 and $\delta(G) \geq 2$, then G has no corner.*

Proof. Suppose $N(x) \subseteq N[u]$ with $u \neq x$. Since $\deg x \geq 2$ we may pick distinct $a, b \in N(x)$. If $u \notin \{a, b\}$ then $a, b \in N(u)$ and $x a u b x$ is a 4-cycle. If $u = a$ then $b \in N(x) \cap N[a]$ and $b \neq a$, so $ab \in E(G)$ and $x a b x$ is a triangle. Either way the girth is at most 4. $\square$

Lemma 3.4. *If H is a retract of G with $|V(H)| = |V(G)| - 1$, then the vertex of $V(G) \setminus V(H)$ is a corner of G . Consequently a corner-free graph has no retract of order $|V(G)| - 1$, and, since retractions compose, none is reached by a chain of retractions either.*

Proof. Let $V(G) \setminus V(H) = \{v\}$ and let f witness the retraction. For $y \in N(v)$ we have $y \in V(H)$, so $f(y) = y$, and $vy \in E(G)$ forces $f(v)y \in E(H)$ or $f(v) = y$; either way $y \in N[f(v)]$. Hence $N(v) \subseteq N[f(v)]$ with $f(v) \neq v$, so v is a corner. For the last statement, a chain of retractions from order $|V(G)|$ to order $|V(G)| - 1$ contains a step that removes exactly one vertex, and a composite of retractions is a retraction. $\square$

Theorem 3.5. *$c(W_1) = c(W_2) = 4$, and neither W_1 nor W_2 has any retract of order 19. Since $M_4 = 19$ and $|V(W_i)| = 20 = M_4 + 1$, the case $k = 4$ of the question quoted in Section 1 is false.*

Proof. Both graphs have girth 5 and are 4-regular, so $c(W_i) \geq \delta(W_i) = 4$ by Theorem 3.1. For the upper bound apply Lemma 3.2 with

$$S = \{0, 14, 16\} \text{ in } W_1, \quad S = \{0, 2, 5\} \text{ in } W_2.$$

In W_1 , $|N[S]| = 14$ and $R = \{2, 5, 11, 12, 15, 19\}$ induces exactly the five edges 2-12, 2-19, 5-15, 5-19, 11-19: a tree on six vertices. In W_2 , $|N[S]| = 13$ and $R = \{1, 3, 4, 7, 10, 12, 19\}$ induces exactly the six edges 1-7, 1-12, 3-10, 4-10, 4-12, 7-19, which form the path 3 10 4 12 1 7 19. Both remainders are forests, so $c(W_i) \leq 4$ and therefore $c(W_i) = 4$; that is, each W_i is 4-cop-win in the sense of [4]. By Lemma 3.3 neither graph has a corner, so by Lemma 3.4 neither has a retract of order 19 – a fortiori none that is 4-cop-win. Since $M_4 + 1 = 20 = |V(W_i)|$, the assertion of the question fails at $k = 4$. $\square$

4. SCOPE

We do not classify the 4-cop-win graphs on 20 vertices: others may exist, of any girth, and no minimality or uniqueness is claimed. $M_4 = 19$ is used, not verified; it is the corollary of [4] that makes 20 the relevant order. Only $k = 4$ of the question is settled: $k = 2$ is refuted in [4] itself, in the very paragraph quoted, by C_5 and C_4 ; $k = 3$ is true there by the classification immediately preceding that paragraph; and $k \geq 5$ is untouched, since M_5 is unknown and nothing here bounds it beyond the $M_5 \geq 20$ already implied by $M_4 = 19$.

5. VERIFICATION

The numerical claims above were re-derived by an accompanying program, `verify.py`, which uses the Python standard library only, in exact integer arithmetic and with no randomness; a transcript of its run is in `verify.output.txt`, and it reports 51 checks, all passing. From the graph6 strings and the adjacency tables of Section 2, transcribed into it as literals, it decodes each string with its own bit-unpacker, re-encodes it, and checks the decoded graph against the adjacency table printed beside it, so the two encodings are cross-checked against each other; it recomputes the order, the size, the degrees, connectivity, the triangle count, $\max_{a \neq b} |N(a) \cap N(b)|$ and the girth; it finds 0 corners over all $20 \cdot 19 = 380$ ordered pairs of each graph; it re-checks the two forest certificates, and all $\binom{20}{3} = 1140$ three-element sets; it computes $|\text{Aut}(W_1)| = 96$ and $|\text{Aut}(W_2)| = 20$, the check on the identification with Meringer’s census entries; and it decides $c(W_1) = c(W_2) = 4$ outright with a bitmask least-fixed-point solver for the k -cop game, so that the value does not rest on Theorem 3.1 or Lemma 3.2.

That solver is controlled on published cop numbers not our own: it returns 1 on P_5 , 2 on C_4 , 2 on C_5 , 3 on the Petersen graph and 4 on the Robertson graph [3], the last two read from the graph6 strings

```
Petersen = IhA@GUAo
Robertson = R???C@_E?KI_I`bGCI?f?ocAoGAQ0?
```

which the program also confirms to be cubic of order 10 and girth 5, and 4-regular of order 19 and girth 5, respectively. The program generates no

graphs, so it does not re-derive the census of [2], and it does not verify $M_4 = 19$.

REFERENCES

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- [4] J. Turcotte and S. Yvon, *4-cop-win graphs have at least 19 vertices*, Discrete Appl. Math. **301** (2021), 74–98. doi:10.1016/j.dam.2021.05.012; arXiv:2006.02998v2. The question refuted here is quoted verbatim in Section 1 from the e-print; it is an unnumbered question in the discussion, distinct from the numbered Conjecture of that paper.